

Mid sem 2026
Biophysics (BIO215)

Total Marks:30

Date: 26.02.2026

Time: 1 Hour

Section A(1 mark each)

1. A researcher is modeling a human protein using a bacterial template that shares 42% sequence identity. During sequence alignment, she observes multiple long insertions in the human protein that are absent in the template. After model building, the core region looks stable, but certain surface regions show poor geometry and steric clashes. What is the most likely reason for the reduced modeling accuracy?

- A) The template structure has high X-ray resolution
- B) Large insertions in the target sequence relative to the template**
- C) Conserved secondary structure elements
- D) High overall sequence identity

2. Which conformational change would be **forbidden** according to Ramachandran plot steric constraints?

- A. $\phi = -60^\circ$, $\psi = -45^\circ$
- B. $\phi = -120^\circ$, $\psi = 120^\circ$
- C. $\phi = +80^\circ$, $\psi = 0^\circ$**
- D. $\phi = -75^\circ$, $\psi = +140^\circ$

3. Is there anything wrong in the given pdb file snapshot opened in a regular text editor (consider the snapshot to be representative of the full pdb file) :

- a. The repeated residue numbers like 142, 143 should not be occurring multiple times, as each amino acid has a fixed residue number for where it occurs in the protein
- b. Problem in the coordinate system**
- c. Information about secondary structure is not present
- d. Mismatch in atom number and residue number

ATOM	1	N	GLU	A	142	10.678	-3.315	0	1.00	28.29	N
ATOM	2	CA	GLU	A	142	11.410	-3.090	0	1.00	28.47	C
ATOM	3	C	GLU	A	142	12.864	-3.594	0	1.00	28.65	C
ATOM	4	O	GLU	A	142	13.794	-2.926	0	1.00	28.78	O
ATOM	5	CB	GLU	A	142	10.678	-3.765	0	1.00	28.28	C
ATOM	6	N	GLN	A	143	13.948	-4.772	0	1.00	28.72	N
ATOM	7	CA	GLN	A	143	14.383	-5.342	0	1.00	28.65	C
ATOM	8	C	GLN	A	143	15.126	-4.634	0	1.00	28.79	C
ATOM	9	O	GLN	A	143	16.347	-4.465	0	1.00	29.11	O
ATOM	10	CB	GLN	A	143	14.294	-6.850	0	1.00	28.63	C
ATOM	11	N	ASP	A	144	14.396	-4.226	0	1.00	28.67	N
ATOM	12	CA	ASP	A	144	14.971	-3.428	0	1.00	28.52	C
ATOM	13	C	ASP	A	144	15.368	-2.085	0	1.00	28.65	C
ATOM	14	O	ASP	A	144	16.275	-1.403	0	1.00	28.43	O
ATOM	15	CB	ASP	A	144	14.001	-3.355	0	1.00	28.38	C
ATOM	16	N	ASP	A	145	14.657	-1.469	0	1.00	28.88	N
ATOM	17	CA	ASP	A	145	15.036	-0.189	0	1.00	29.28	C
ATOM	18	C	ASP	A	145	16.422	-0.288	0	1.00	28.95	C

4. A researcher introduces multiple **glycine residues** into the core of a globular protein. After modification, the protein becomes **less stable and more flexible**. Which of the following best explains this observation?
- A) Glycine increases steric hindrance, destabilizing folding
 - B) Glycine lacks a side chain, increasing conformational freedom and reducing stability**
 - C) Glycine forms extra hydrogen bonds that disrupt secondary structure
 - D) Glycine increases hydrophobic packing in the protein core
5. Being a mad scientist, you wanted to make a Genetically modified human, capable of breathing under water. You hypothesized, instead of making gills as in fishes, what if I introduce proteins that will undergo rapid conformational changes in presence of water and salt concentration. Which functional statement about water is MOST relevant to protein folding?
- a. Water only acts as a dielectric and has no role in hydrophobic collapse rather drives hydrophilic residues in the core and hydrophobic on the surface.
 - b. Water drives the hydrophobic effect, pushing nonpolar residues into the protein core**
 - c. Water forms covalent bonds with side chains to stabilize folding
 - d. Water prevents hydrogen bonds from forming in proteins, leading to greater flexibility required for conformational changes.
6. Bacteriorhodopsin spans the membrane via multiple α -helices. What property enables membrane insertion?
- A) Polar surface residues
 - B) Repeating glycine residues
 - C) Disulphide Bonds
 - D) Hydrophobic amino acids facing lipid bilayer**
7. Two neutral noble gas atoms are placed far apart in the gas phase. As the distance between them increases significantly, all strong bonding interactions disappear. However, a very weak attractive force still remains between them. Which interaction is responsible for this residual attraction at large intermolecular separation?
- A) Overlap of valence orbitals

B) Permanent dipole–dipole interaction

C) Instantaneous dipole–induced dipole interaction

D) Ionic electrostatic attraction

8. Match the structure with its approximate ϕ , ψ region:

Column I	Column II
1. α R helix	a. $\phi \approx -60^\circ$, $\psi \approx -45^\circ$
2. β -sheet	b. $\phi \approx -120^\circ$, $\psi \approx 120^\circ$
3. α L helix	c. $\phi \approx +60^\circ$, $\psi \approx +45^\circ$

Correct matching:

A) 1-a, 2-b, 3-c

B) 1-b, 2-a, 3-c

C) 1-c, 2-b, 3-a

D) 1-a, 2-c, 3-b

9. Which statement correctly compares typical lengths?

A) α -helices are usually shorter than β -strands

B) β -strands are usually longer than helices

C) α -helices can range 5–40 residues, β -strands 5–10 residues

D) Both have identical length ranges

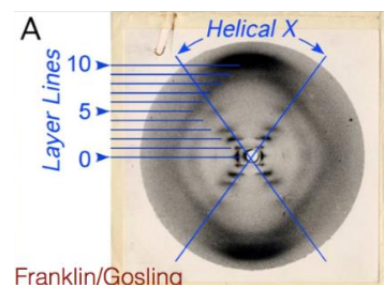
10. The strong reflections at regular intervals in Franklin's diffraction pattern indicated:

a. Random coil structure

b. Periodic helical structure

c. Protein contamination

d. Covalent crosslinking



Questions (2 marks each)

1. Compare Pauling's triple helix model with the accepted double helix model
 - **Description of Pauling's model – 0.5 marks**
Pauling proposed a triple helical structure with three strands and phosphate groups positioned at the center. This arrangement caused electrostatic repulsion due to negatively charged phosphates.
 - **Description of correct model – 0.5 marks**
The accepted double helix model consists of two antiparallel strands with phosphate backbones on the outside and nitrogen bases inside. Hydrogen bonding occurs between complementary bases (A–T, G–C).
 - **Key differences (phosphate position, bonding, stability) – 1 mark**
The major differences include the number of strands, placement of phosphate groups, and stability due to hydrogen bonding and reduced electrostatic repulsion.

2. Explain why engineered “chameleon sequences” invalidate local-window secondary prediction assumptions
 - **Description of prediction assumption – 0.5 marks**
Early secondary structure prediction methods (e.g., Chou–Fasman) assume that secondary structure is determined mainly by local amino acid sequence, using a sliding window (~13–17 residues) without considering long-range interactions.
 - **Description of chameleon sequence – 0.5 marks**
Chameleon sequences are peptide segments that can adopt different secondary structures (α -helix or β -strand) depending on their structural or environmental context, even though their amino acid sequence remains the same.
 - **Key reason for prediction failure – 1 mark**
Chameleon sequences demonstrate that non-local interactions and tertiary structure context influence secondary structure, invalidating the assumption that local sequence alone determines folding. This reduces the reliability of purely local-window prediction methods.

3. In a phylogenetic tree, gene A duplicates into A1 and A2. Later, speciation produces species X and Y. We observe: 1. X-A1 and Y-A1 2. X-A2 and Y-A2
 - a. What is the relationship between X-A1 and Y-A1? **(0.5 mark): Orthologs**
 - b. What is the relationship between X-A1 and X-A2? **(0.5 mark): Paralogs**
 - c. Why are orthologs generally better templates for homology modelling than paralogs? **(1 mark)**

Ans: Orthologs usually retain similar function after speciation**(0.5 mark)**. Similar function implies conserved structure, making structural modeling more reliable.**(0.5 mark)**

4. Why are hydrogen bonds in aqueous proteins considered entropy-driven rather than purely enthalpy-driven?

Ans: [bond exchange (protein-water $\rightarrow$ protein-protein)] – 1 mark

[Water entropy increase as driving force] – 1 mark

Forming an internal hydrogen bond often replaces hydrogen bonds previously formed with water. The enthalpy change is small because bonds are exchanged rather than created. The main driving force is the release of ordered water molecules, increasing solvent entropy.

5. Why can burying a charged residue inside a protein core be energetically catastrophic, even though electrostatic interactions are stronger inside proteins?

Ans: The inside of a protein is mostly nonpolar (hydrophobic). Charged residues prefer water because water stabilizes charges(0.5 mark). In water, charged residues are stabilized by strong interactions with polar water molecules. When buried inside the protein, they lose this stabilization(0.5 mark). Unless the charge forms a salt bridge or hydrogen bond inside the protein, it remains energetically unfavorable (0.5 mark). Even though electrostatic interactions are stronger inside proteins (low dielectric constant), placing an isolated charge in a nonpolar environment requires a large amount of energy, making it unstable(0.5 mark).

Section c

1. During molecular modeling of a newly designed enzyme, you notice that the folded protein structure shows several steric clashes in the hydrophobic core. Answer the following:

(a) what happens to energy as interatomic distance (w.r.t LJ Potential) $r \rightarrow 0$? (1 mark)

As $r \rightarrow 0$, the repulsive $1/r^{12}$ term dominates, and the energy rises steeply towards $+\infty$, representing steric clashes.

(b) Why are hydrophobic protein cores mainly stabilized by van der Waals interactions?(1 mark)

Hydrophobic cores rely on van der Waals interactions because nonpolar side chains cannot form hydrogen bonds or ionic interactions; instead, they are stabilized by many weak dispersion forces acting cooperatively.

(c) What do you understand by ‘Clashes’? How do van der Waals radii help prevent steric clashes?(2 marks)

Clashes - unfavorable interactions where atoms are too close together; close contacts.

Van der Waals radii represent the effective "size" of atoms. They define a minimum allowed distance between atoms, helping to avoid steric clashes.

(d) Explain how the balance of Lennard–Jones attractive and repulsive forces determines protein packing density. **(1 mark)**

The attractive ($1/r^6$) term pulls atoms together, while the repulsive ($1/r^{12}$) term pushes them apart at very short distances. The balance creates an equilibrium distance, which sets how tightly atoms can pack inside a protein.

2. During an expedition to MARS, you encountered a slime-like life form. You decided to bring it back to earth as the concrete evidence of alien life finally ending the long standing fermi paradox. You stored the slime in an isolated container mimicking the martian environment. Once brought on earth, the slime made contact with earth's environment. You observed that the slime started going through rapid stiffening, while returning back to in martian like conditions. Based on this information:

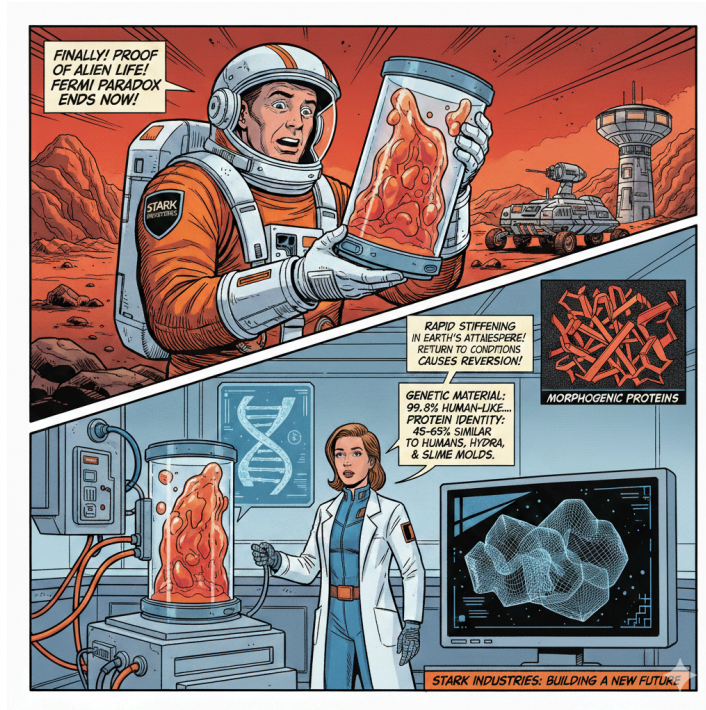
A) Why do you think this rapid morphogenesis occurred? **(0.5 mark)**

Ans A- Environmental changes altering intermolecular interactions, driving rapid conformational changes (or similar idea)

B) Assuming that the alien slime shares the same kind of genetic material as us, what kind of proteins do you believe could make it possible for it to have such powers? Explain **(2 mark)**

Ans B- This behaviour suggests presence of predominantly globular and/or loosely organized fibrous structural proteins whose conformations depend strongly on environmental factors like temperature (-60 to -65 degrees on Mars vs ~10-15 degrees on Earth). Flexible globular proteins can rapidly switch to compact/expanded states in response to environmental changes while fibrous/matrix associated proteins can reversibly associate to form stiff networks (Basically a similar idea for the answer should be presented).

C) You wanted to study the structure of these proteins to see if we can use this alien slime feature to make Xenohumans. With the help of Stark Industries, you were able to secure funding to get the entire proteome sequenced and now you have the sequence of the target protein. Give a brief approach for how you will go about modelling it in silico considering: The protein shares makeup with only 4 other proteins from humans, hydra, some myxomycetes and bats, with identity ranging from 45%-65%. Also state the strategies for evaluating the model. **(2 + 0.5 mark)**



Ans C- MSA -> Identify conserved domains -> Multi-template selection based on highest identity for each domain -> Generate multiple models -> Evaluation using DOPE/RMSD & Ramachandran analysis -> Loop refinement if need be. (This is the core idea of the answer, 1 mark for stating at least 2 evaluation criterias).